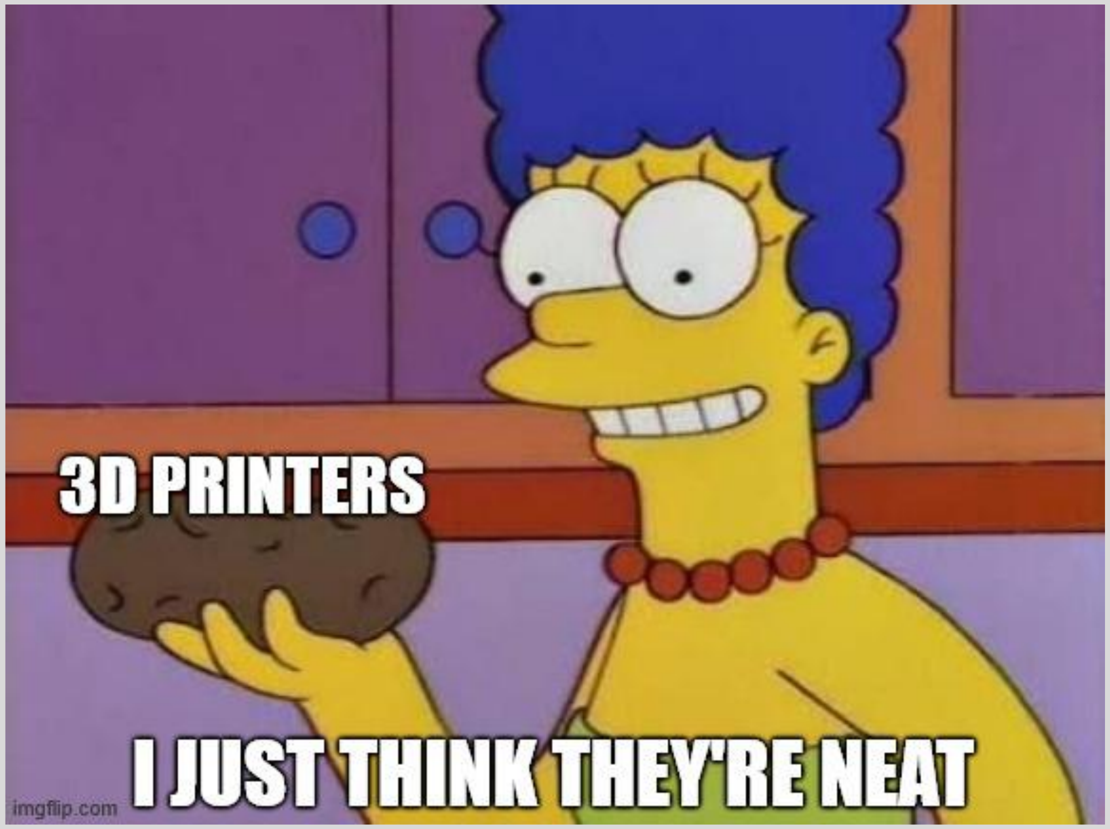
FFF Printing Fundamentals

Part 4: Advantages

An Amazing Tool

- Playground for **creative design** and **creative problem solving**.
- Massive utility for cheap.
- Make replacement parts to fix and repair.
- Bring your ideas to life.
- Fast turnaround time.
- Low barrier to entry.
- Practical for hobbyists, students, and professionals.

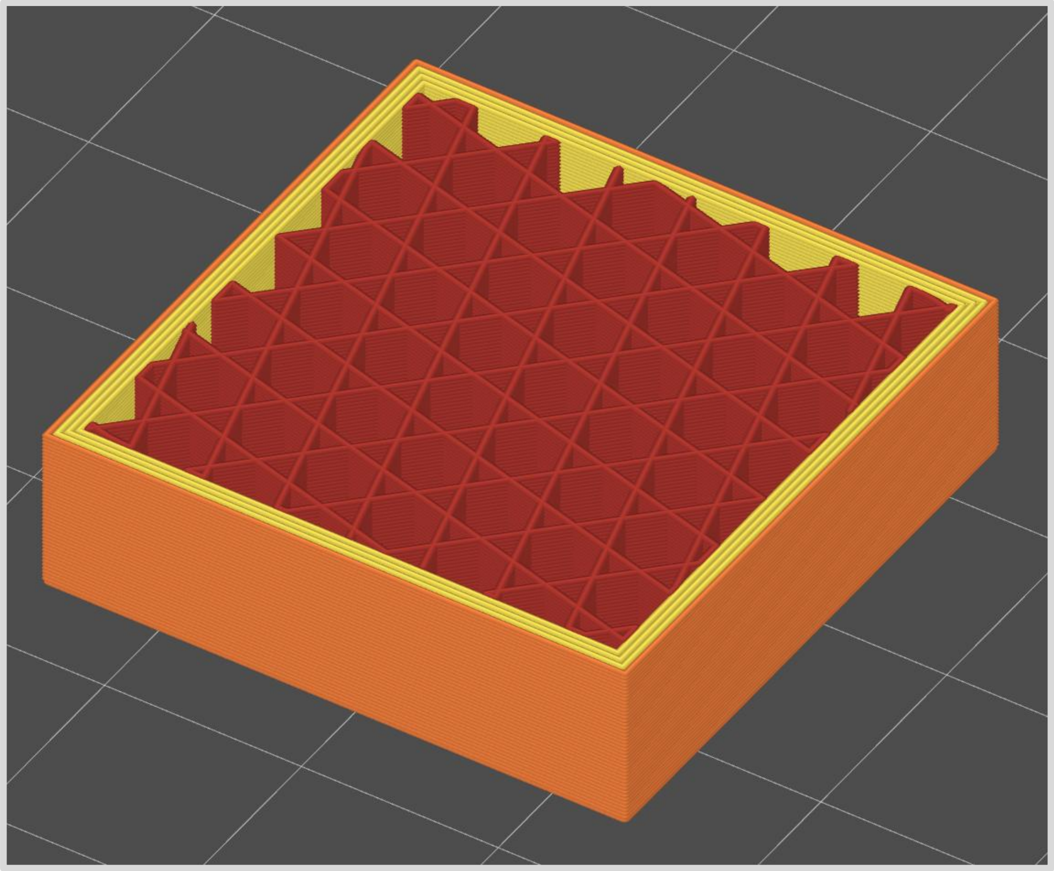


An Amazing Community

- RepRap festivals, maker fairs, local clubs and maker spaces, …
- People share cool projects and ideas. Examples include:
  - *Full spectrum printing*
  - *BumpMesh.com*
  - *Voron*
- YouTube makers with educational content. Examples include:
  - *CNC Kitchen*
  - *Maker's Muse*
  - *JanTec Engineering*
  - *3D Printing Nerd*
  - *Teaching Tech (retired)*

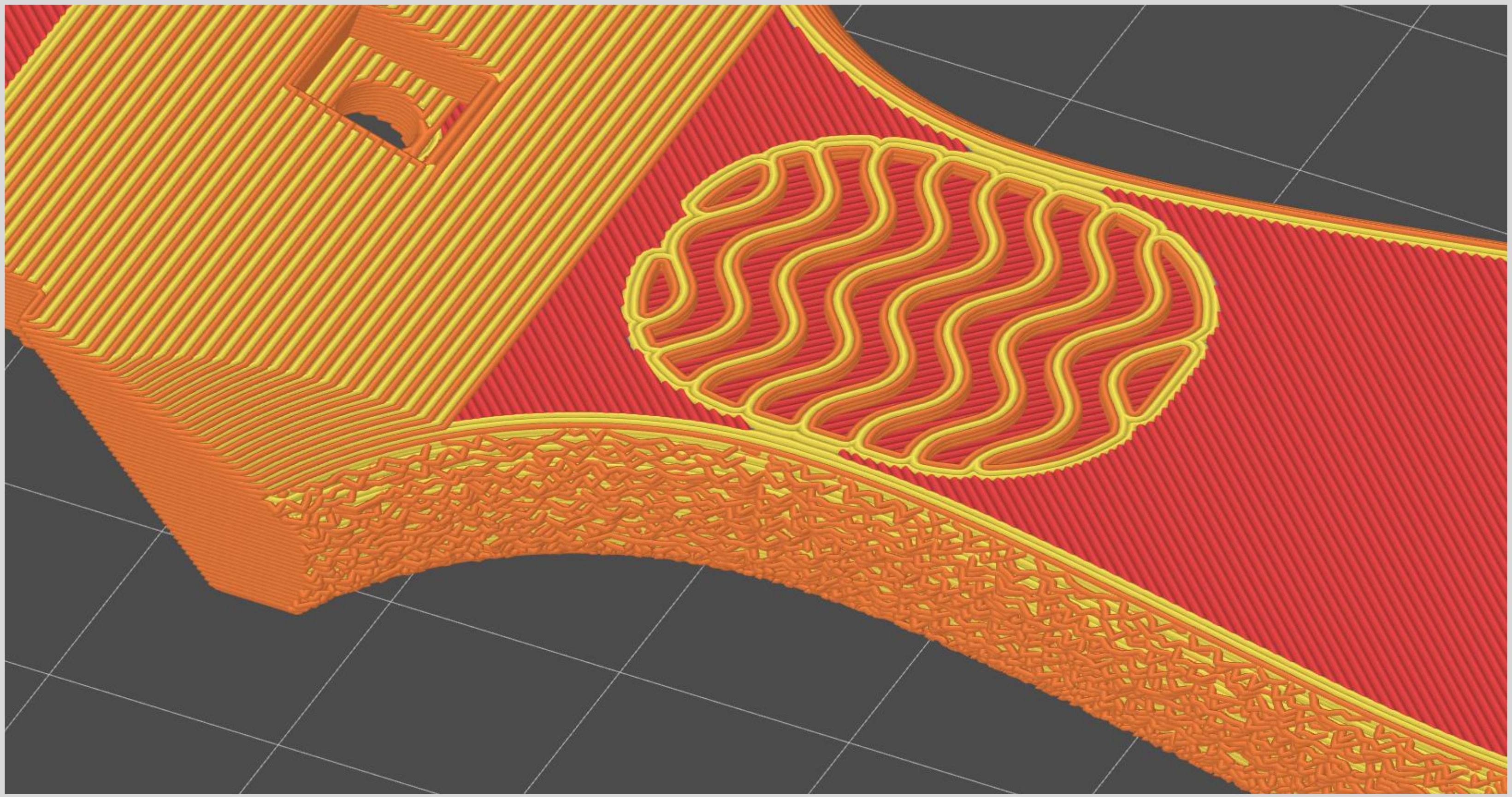
FFF Advantages

- Make lots of shapes with tiny “building blocks.”
  - Typical extrusions are
    0.2 mm tall (1-2 pieces of paper)
    0.4 mm wide.
- Not restricted to uniform, solid objects.
  - Easily create internal geometry.
  - Great at making hollow objects.



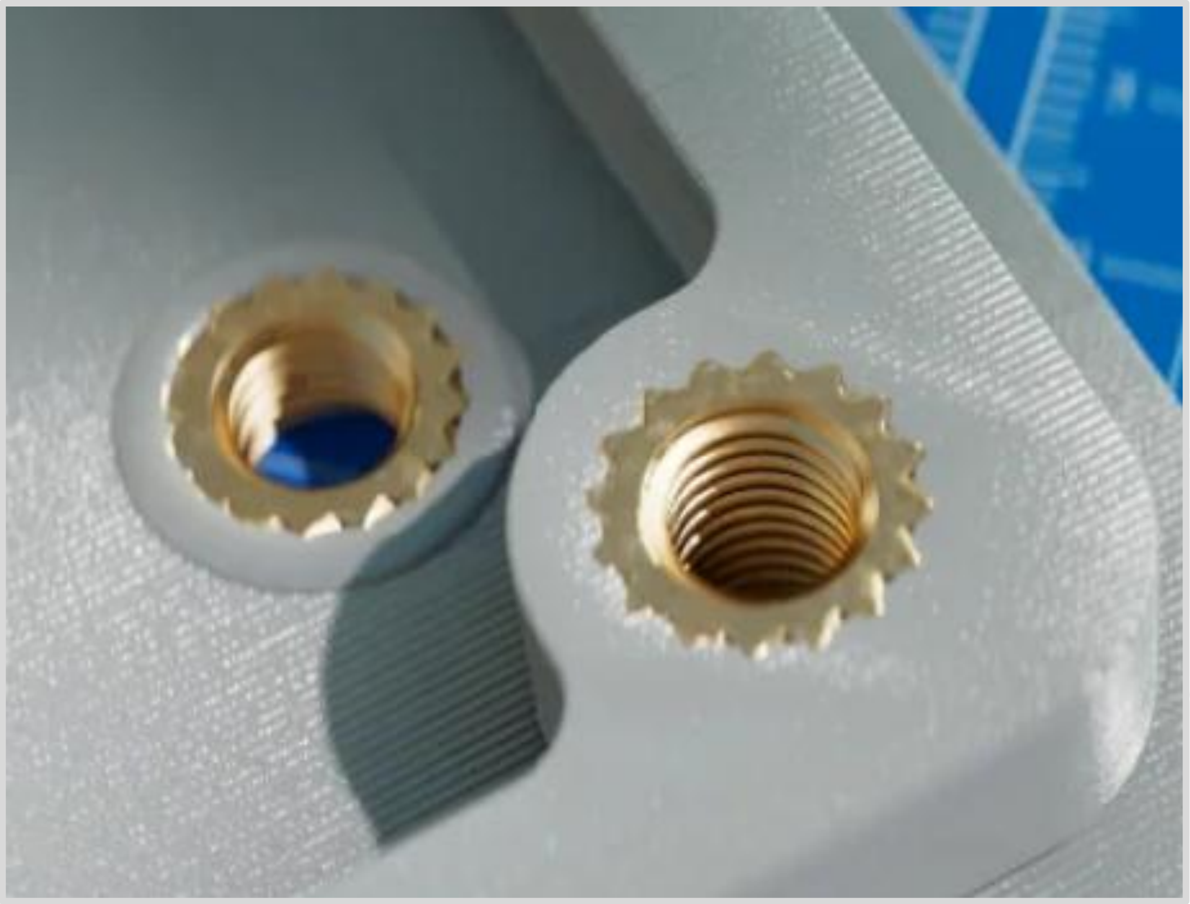
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Threaded anchor for screws. Why?
- Plastic is soft →
  Screws easily wear out plastic holes.
- Install with a hot soldering iron.
- Locks in place
  when it cools.
- Very durable.



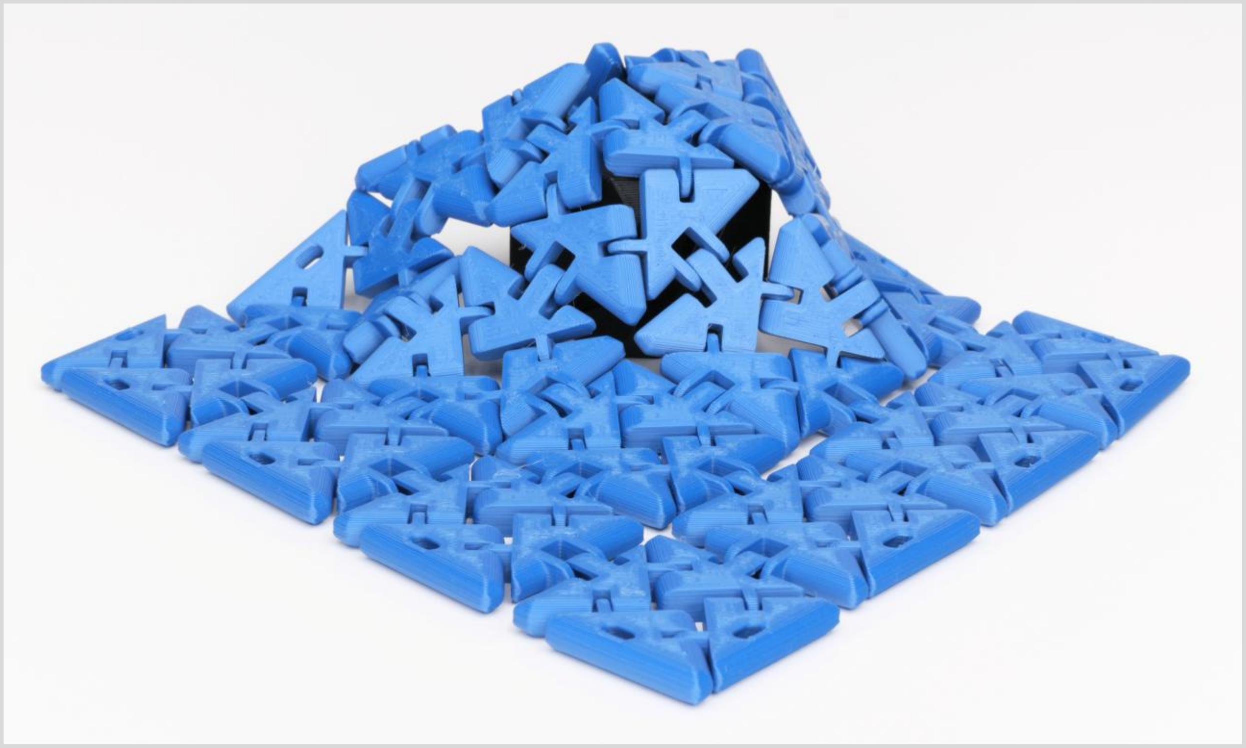
(Pictures from CNC Kitchen Store)

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
  - Nuts and fasteners.
  - Magnets.
  - Weights.
  - Reinforcing bar or plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
  - Need to be at the printer at the right layer.

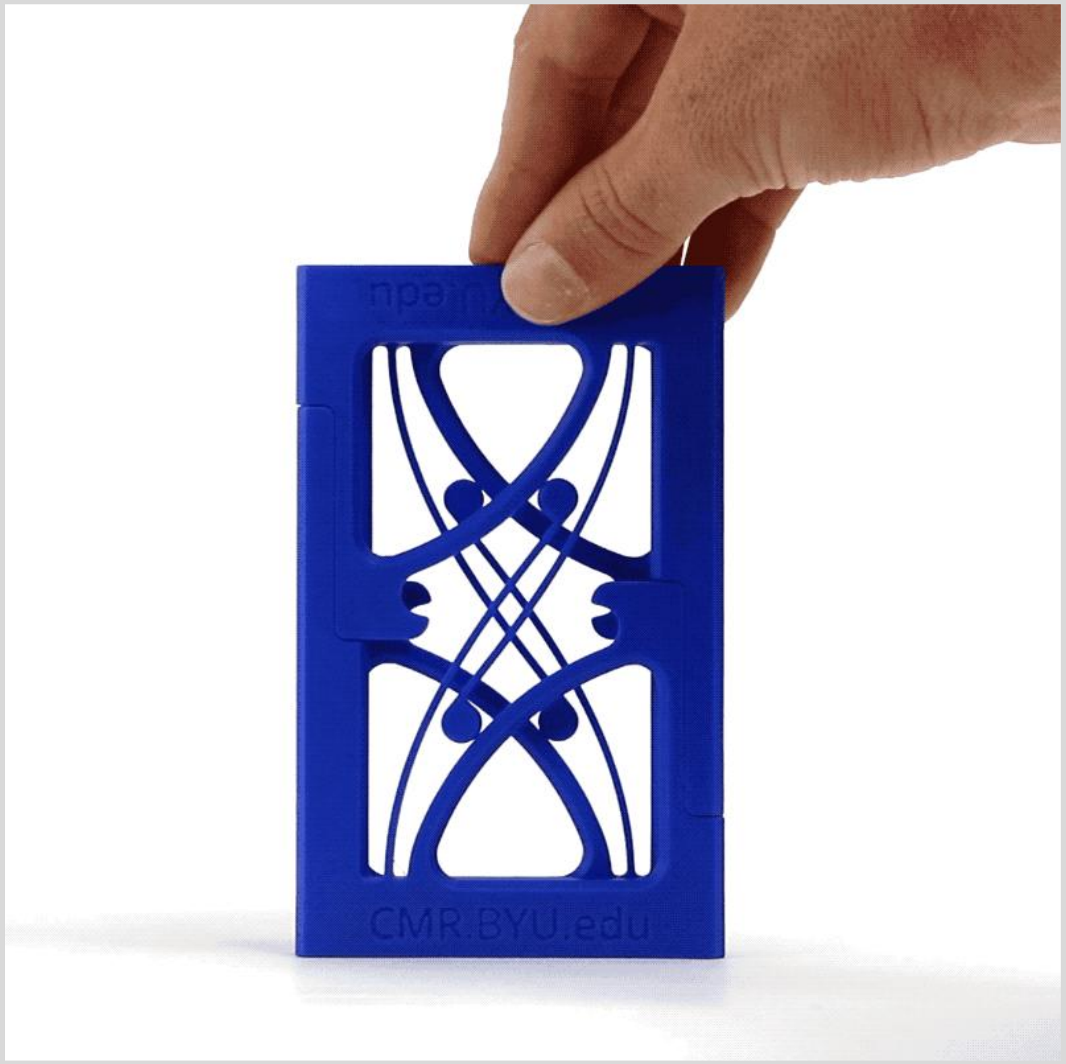
Print-in-Place (PIP)

- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by printables.com/@Flying Gyroscope



Functional Flexing

- Springs and **compliant mechanisms**.
- Features and shapes that function by bending and flexing.
- Elastic, springy materials.
  - No permanent deformation.
- Remember part strength and layer lines.

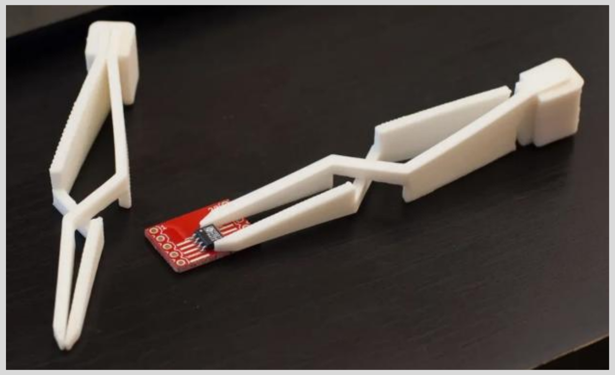


(Animation, Constant Force Mechanism by BYU CMR)

Springs by VC Design

Clip by AE

Cross tweezer by joo



Living hinge by DrJones